

### 3. A map that stretches the short side of the domain

This degree 1 map takes

$$R_1 \times R_2 \times R_3 \times R_4$$

onto

$$\lambda R_1 \times \lambda^{-3} R_2 \times \lambda R_3 \times \lambda^{-1} R_4.$$

The constant  $\lambda$  is allowed to take any value which is at least 1, smaller than  $(R_2/R_1)^{1/4}$ , and smaller than  $(R_4/R_3)^{1/2}$ .

Construction:

Let  $I$  be a quasi-isometric embedding of  $\lambda^{-2} R_2 \times \lambda^2 R_3 \times R_4$  into  $R_2 \times R_3 \times R_4$ .

Let  $U$  be the union of  $R_1 \times \text{image}(I)$  with  $\{0\} \times R_2 \times R_3 \times R_4$ .

Step 1. There is a Lipshitz retraction of  $R$  onto  $U$ .

Step 2. There is a retraction of  $U$  onto  $R_1 \times \text{image}(I)$ , which has 2-dilation bounded by  $\lambda^2$ . Moreover, at any point of  $U$ , either this map is 2-contracting, or else the derivative of the retraction maps into the 3-plane spanned by  $\partial/\partial x_2, \partial/\partial x_3$ , and  $\partial/\partial x_4$ .

Step 3. There is a quasi-isometric mapping from  $R_1 \times \text{image}(I)$  to  $R_1 \times \lambda^{-2} R_2 \times \lambda^2 R_3 \times R_4$ , which preserves the coordinate  $x_1$ .

Step 4. Compose with the linear map from  $R_1 \times \lambda^{-2} R_2 \times \lambda^2 R_3 \times R_4$  to  $\lambda R_1 \times \lambda^{-3} R_2 \times \lambda R_3 \times \lambda^{-1} R_4$ . This linear map is 2-contracting. Moreover, its restriction to the 3-plane  $x_1 = 0$  has 2-dilation  $\lambda^{-2}$ . Therefore, the composition of all four maps is 2-contracting.

Technical remark: If we take  $I$  to be a quasi-isometric embedding of  $\lambda^{-2} R_2 \times \lambda R_3 \times \lambda R_4$  into  $R_2 \times R_3 \times R_4$ , then the same argument gives us a map from  $R$  to  $\lambda R_1 \times \lambda^{-3} R_2 \times R_3 \times R_4$ . By an argument interpolating between these cases, we get a map to  $\lambda R_1 \times \lambda^{-3} R_2 \times A \times B$ , where  $\lambda^4 < (R_2/R_1)$ ,  $R_3 < A < (R_3 R_4)^{1/2}$  and  $R_3 R_4 = AB$ .

### 4. Pinching Map

This degree 1 map takes

$$R_1 \times R_2 \times R_3 \times R_4$$

onto